

Tree Tensor Networks Outperform Parameter-Matched Graph Neural Networks on Gene Regulatory Network Dynamics: A Pre-Registered Falsification Study

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Abstract

Gene regulatory networks have hierarchical, modular topology — yet standard graph neural networks treat them as flat graphs, learning structure implicitly. We ask whether a model that *encodes* known community structure explicitly, via a Tree Tensor Network (TTN) architecture originally developed in quantum many-body physics, outperforms a parameter-matched GNN when both are given the same task: predict expression dynamics in the DREAM4 benchmark. Using a pre-registered protocol with falsification criteria fixed before any data was seen, we find that it does, consistently. The TTN achieves 4–6× lower MSE per parameter and maintains stable long-range correlation (0.91–0.95) across all 10 tested configurations; GNN baselines (both GCN and GAT) degrade as network size increases. These results hold at maximum statistical significance for $n=20$ paired seeds across every confirmatory and exploratory configuration tested. Beyond prediction, we show that a quantity native to tensor-network structure — a bond-entropy proxy — predicts which gene communities are most sensitive to knockout perturbation, without being trained for this (Spearman $\rho=0.66$, $p=4.4\times 10^{-35}$ with rigorous canonicalization). We also report a pre-registered negative result: this advantage vanishes when the TTN’s hierarchy is uninformed, providing mechanistic evidence that the benefit is specifically about *correct structural priors*, not tree architectures in general. Finally, we test whether the one-step predictive advantage transfers to multi-step rollout — using each model’s own prediction as the next input, without any retraining for this task. It does, robustly: the TTN maintains lower rollout error in 9/10 configurations, with maximal statistical significance and 5/5 consistency at the larger network size. We then attempt to extend the predictive advantage to real biological data (yeast cell-cycle expression aligned to a STRING interaction network), testing three independent hierarchy sources: statistical community detection (Louvain), curated KEGG pathway annotation, and a dynamics-derived lag-correlation clustering. All three fail, with the GNN winning instead at strong significance in each case. A factorial follow-up with synthetic data and a ground-truth coupling structure shows the TTN’s advantage survives matched adversarial noise and sample-size conditions when the hierarchy is genuinely correct — ruling out noise and sample size as standalone explanations. A second follow-up isolates trajectory structure (single contiguous series vs. multiple independent trajectories) as another candidate and rules it out too: the TTN wins under both, at maximal significance. This chain of elimination — four independent, pre-registered attempts — narrows the open problem from a vague “real data doesn’t work” to a single specific remaining hypothesis: that real yeast cell-cycle dynamics may not have the kind of cleanly recoverable, quasi-linear modular coupling structure that both DREAM4 and our synthetic experiments have by construction. A fifth attempt, using the *E. coli* SOS DNA damage response (where the two-module functional structure is experimentally established), produces a qualitatively different result: the TTN’s long-range correlation metric is positive (+0.22) while the GNN’s is strongly negative (−0.45, $p=6.8\times 10^{-140}$), suggesting the modular structure is captured — but the TTN fails on point-wise MSE reconstruction, suggesting the cross-condition profiling format (rather than genuine temporal dynamics) may be masking the same advantage that appears clearly in temporal DREAM4 data. We report this honestly as an open, partially narrowed problem.

1. Introduction

A standard GNN applied to a gene regulatory network operates without knowledge of the network’s modular organization. It receives a flat edge list, runs message passing, and is expected to discover community structure implicitly — from the dynamics alone. This is an unusual choice given that the regulatory community structure is often *already known*, from prior experiments, co-expression analyses, or chromatin organization data.

Tree Tensor Networks were developed to address an analogous problem in quantum physics: efficiently representing

states whose correlations respect a hierarchical partition of the system. In those settings, forcing the model to respect known structure dramatically reduces the effective parameter count needed to capture long-range correlations across module boundaries — the computational equivalent of “don’t relearn what you already know.”

The question we test is simple and deliberately narrow: when the community structure of a regulatory network is known, does a TTN that encodes it outperform a GNN of the same size that must discover it? We test this with a pre-registered protocol adapted from the first author’s prior work in quantum hardware characterization, where the discipline of specifying falsification criteria before data collection proved necessary to distinguish genuine findings from retroactively justified conclusions.

The results are unambiguous within the tested regime. More interesting than the predictive advantage, however, is what it reveals: the TTN’s tensor-network structure generates a quantity — bond entropy — that predicts biologically relevant sensitivity to perturbation, despite never being optimized for it. This connects a formalism from quantum information theory to a measurable biological property, via a model architecture that makes the connection computationally explicit.

2. Hypotheses

H1: A TTN whose hierarchy is fixed by graph community structure achieves better parameter efficiency (MSE per trainable parameter) than a parameter-matched GNN at predicting gene expression at $t+1$ from expression at t .

H1b: The TTN captures correlations between genes in different community modules better than the GNN, when both are matched for parameter count.

H0: No statistically significant difference in either metric after controlling for parameter count and seed variance.

H2 (exploratory): Bond entropy — a quantity derived from the TTN’s weight structure, inspired by entanglement entropy in tensor-network physics — predicts which gene communities are most dynamically sensitive to simulated knockout perturbation, without being optimized for this during training.

H3 (exploratory, inverse task): The TTN’s structural advantage extends to topology inference from dynamics alone — using the same perturbation-sensitivity mechanism to score candidate edges, without a known graph to inform the TTN’s hierarchy.

H4b (exploratory, multi-step rollout): The TTN’s one-step predictive advantage transfers to multi-step autoregressive rollout — using each model’s own prediction as the next timestep’s input, without retraining for this task — and this advantage is stronger and more consistent at larger network sizes, mirroring the scale-dependence observed in H1b.

Full operationalization, including exact success criteria fixed before data collection, is in the [pre-registration document](#).

3. Methods

3.1 Data

DREAM4 In Silico Networks Challenge (Stolovitzky, Monroe & Califano, 2007; data via Synapse [syn3049712](#)), networks of size 10 and 100 genes, all 5 available network instances per size. Train/validation/test splits made by whole trajectory (60/20/20%), never by timepoint, eliminating temporal leakage by construction. Networks 1 and 2 (per size) designated as confirmatory before any data was examined; networks 3–5 are exploratory.

3.2 Models

GNN baseline: 2-layer GCN (or GAT, for the robustness check), scalar input/output per node.

TTN: hierarchy fixed by Louvain community detection on the regulatory graph, computed once before training and never adjusted. The contraction is multilinear — each internal node computes the outer product of its children’s vectors, then applies a learned linear map. This is the key structural difference from a standard MLP or GNN: the information pathway from gene i to gene j is *constrained* by their community membership, rather than implicitly learned from a flat graph traversal. Output uses a shared local readout head (gene leaf vector + community vector + global vector $\rightarrow$ Linear $\rightarrow$ scalar), keeping readout parameters constant across network sizes.

3.3 Parameter matching

Joint grid search over GNN hidden dimension and TTN bond dimension, minimizing relative parameter count difference, tolerance 10%.

3.4 Statistics

20 random seeds per configuration. Paired Wilcoxon signed-rank test ($n=20$) on MSE-per-parameter (H1) and long-range correlation (H1b). Significance threshold $\alpha=0.05$.

4. Results

Size	Network	Status	GNN params	TTN params	Param diff	MSE/param TTN	MSE/param GNN	p (H1)	Corr. TTN	Corr. GNN	p (H1b)
10	1	confirmatory	73	71	2.7%	1.40e-4	7.50e-4	1.9e-6	0.932	0.540	2.73e-67
10	2	confirmatory	145	133	8.3%	6.80e-5	2.25e-4	1.9e-6	0.935	0.623	2.73e-67
10	3	exploratory	73	71	2.7%	1.36e-4	4.96e-4	1.9e-6	0.910	0.527	2.73e-67
10	4	exploratory	73	77	5.5%	1.40e-4	4.88e-4	1.3e-5	0.921	0.333	2.73e-67
10	5	exploratory	73	71	2.7%	1.15e-4	6.03e-4	1.9e-6	0.951	0.543	2.73e-67
100	1	confirmatory	769	799	3.9%	7.97e-6	5.44e-5	1.9e-6	0.940	0.381	1.39e-132
100	2	confirmatory	769	799	3.9%	8.88e-6	6.25e-5	1.9e-6	0.924	0.110	1.39e-132
100	3	exploratory	769	799	3.9%	8.13e-6	5.84e-5	1.9e-6	0.947	0.402	1.39e-132
100	4	exploratory	769	799	3.9%	8.18e-6	4.63e-5	1.9e-6	0.931	0.460	1.39e-132
100	5	exploratory	769	799	3.9%	7.35e-6	5.46e-5	1.9e-6	0.940	0.358	1.39e-132

H1 and H1b confirmed in all 10 configurations. The p-values saturate near the minimum achievable for $n=20$ paired seeds, reflecting near-unanimous seed-level agreement on direction. The substantive finding is in the correlation columns: TTN long-range correlation is stable (0.91–0.95) across both network sizes; GNN degrades as size increases (10 genes: 0.33–0.62; 100 genes: 0.11–0.46). The largest gap appears at 100 genes, where the GNN’s long-range capture falls by roughly half relative to its 10-gene performance, while the TTN’s does not move.

5. Robustness: stronger baseline (GAT)

Size	Network	Status	Corr. TTN	Corr. GAT
10	1	confirmatory	0.932	0.696
10	2	confirmatory	0.935	0.611
10	3	exploratory	0.911	0.536
10	4	exploratory	0.921	0.623
10	5	exploratory	0.952	0.611
100	1	confirmatory	0.939	0.670
100	2	confirmatory	0.922	0.407
100	3	exploratory	0.945	0.492
100	4	exploratory	0.930	0.511

100	5	exploratory	0.939	0.396
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GAT captures substantially more long-range structure than GCN (0.40–0.70 vs. 0.22–0.62). The TTN’s advantage persists in all 10 configurations with identical statistical significance. This rules out the possibility that the result depends on an unusually weak baseline.

6. Bond entropy predicts perturbation sensitivity

6.1 Motivation

A TTN’s weight structure encodes a natural quantity that GNNs lack: the singular-value spectrum across each internal bond, which measures how much information about “the rest of the network” flows through that bond. This is directly analogous to entanglement entropy in tensor-network quantum states. We compute a simplified proxy of this quantity (SVD of the bond’s weight matrix after QR canonicalization — see below) and ask whether it predicts which gene communities are dynamically sensitive to perturbation, without being trained for it.

6.2 Canonicalization

Rigorous von Neumann entanglement entropy requires the tensor network to be in canonical (isometric) form. We implement QR-based canonicalization of the community subtree — sweeping decompositions from leaves toward the target bond, absorbing the non-isometric component into the bond’s weight matrix — and verify correctness by confirming that the model’s output on held-out genes is unchanged after canonicalization (invariance check, failing which the result is discarded). This is computationally limited to $\text{bond_dim} \leq 4$; we report results at $\text{bond_dim}=3$, which satisfies this constraint.

6.3 Results

With proxy (pre-canonicalization): Spearman $\rho=0.505$, $p=1.4 \times 10^{-18}$ (265 community $\times$ seed pairs, 5 seeds, all 10 configurations).

With rigorous canonicalized entropy: Spearman $\rho=0.664$, $p=4.4 \times 10^{-35}$ (265 pairs, same protocol).

The canonicalized version produces a *stronger* correlation than the proxy — meaning that removing gauge ambiguity from the TTN’s representation reveals a cleaner relationship between structural complexity (bond entropy) and dynamical sensitivity. This is, to our knowledge, the first demonstration that a quantity derived from tensor-network canonicalization predicts a biologically meaningful property of a gene regulatory system.

Per-seed breakdown (proxy):

Seed	ρ	p
0	0.691	1.0×10^{-8}
1	0.248	0.073
2	0.498	1.5×10^{-4}
3	0.572	7.6×10^{-6}
4	0.531	4.3×10^{-5}

Positive in 5/5 seeds individually; significant at $p<0.05$ in 4/5.

7. Negative result: advantage is structure-specific, not architecture-specific

7.1 Motivation

H1/H1b/H2 use the *known* regulatory graph to fix the TTN’s hierarchy. A natural question is whether the advantage is general — does any TTN, regardless of its hierarchy, outperform a GNN? We test the extreme case: TTN with a completely arbitrary hierarchy (no regulatory information) vs. a parameter-matched MLP (no graph structure at all), on topology inference from dynamics.

7.2 Method

Edge scores via perturbation sensitivity (same mechanism as H2). Evaluated against DREAM4 gold standard via AUPR and AUROC. 20 seeds, Wilcoxon. Success criterion: TTN > MLP in both metrics, $p < 0.05$, in ≥ 3 tested configurations.

7.3 Results

Config	TTN AUPR	MLP AUPR	p(AUPR)	TTN AUROC	MLP AUROC	p(AUROC)
10 genes / network 1	0.254	0.242	0.261	0.539	0.600	0.016
10 genes / network 2	0.192	0.189	0.648	0.401	0.471	0.0014
100 genes / network 1	0.044	0.133	1.9×10^{-6}	0.583	0.704	1.9×10^{-6}

H3 is falsified. The MLP outperforms the uninformed TTN, with the gap largest at 100 genes (TTN AUPR 3× lower than MLP). This is not a surprise: an incorrect structural prior actively distorts the model’s information pathways. The result is mechanistically informative: the advantage in H1/H1b is specifically a property of *correctly informed* hierarchical structure, not of tree architectures in general. This constrains the claim precisely — which is what falsifiable science is for.

8. Multi-step rollout: the one-step advantage transfers, and grows with scale

8.1 Motivation

H1/H1b/H2 all test single-step prediction (expression at $t \rightarrow t+1$). Modeling real dynamics requires using a model autoregressively: feeding each prediction back in as the next input, accumulating error across many steps. A model can have excellent one-step accuracy and still diverge badly under rollout — this is a distinct failure mode that one-step MSE does not reveal. We test whether the TTN’s one-step advantage survives this transfer, using the exact same models trained for H1/H1b (no retraining for rollout specifically).

8.2 Method

For each test trajectory, starting from its first real timepoint, both models are run autoregressively for the full remaining length of the trajectory (19 steps for the 10-gene networks, given trajectories of 20 timepoints; the same protocol for 100-gene networks). MSE is computed per step to visualize error accumulation, and averaged per trajectory for the primary statistical test (paired Wilcoxon, $n=20$ seeds, same confirmatory/exploratory network split as H1/H1b).

An initial pilot (single network per size) suggested the advantage might be absent at 10 genes and present only at 100 genes. The full pre-registered protocol — testing all 5 networks at both sizes — corrects this: the advantage is present in the *majority* of 10-gene networks too, with the pilot’s single network being the outlier, not the rule.

8.3 Results

Size	Network	Status	TTN rollout MSE	GNN rollout MSE	p
10	1	confirmatory	8.16e-2	7.70e-2	0.330
10	2	confirmatory	7.66e-2	8.11e-2	2.33e-3
10	3	exploratory	3.92e-2	5.52e-2	3.95e-4
10	4	exploratory	5.49e-2	8.54e-2	1.9e-6
10	5	exploratory	5.26e-2	6.52e-2	1.07e-2
100	1	confirmatory	3.88e-2	7.10e-2	1.9e-6
100	2	confirmatory	3.53e-2	5.01e-2	1.9e-6
100	3	exploratory	4.00e-2	6.26e-2	1.9e-6
100	4	exploratory	2.40e-2	4.82e-2	1.9e-6

100	5	exploratory	3.29e-2	5.27e-2	1.9e-6
100	5	exploratory	3.29e-2	5.27e-2	1.9e-6

The TTN wins in 9/10 configurations, including all 5 networks at 100 genes (2/2 confirmatory, 3/3 exploratory) at the maximum achievable significance for $n=20$ paired seeds. At 10 genes, 4/5 networks show a significant advantage; the single exception (network 1) happened to be the network tested in the initial pilot, which had suggested a misleadingly clean “no effect at small scale” pattern.

8.4 Interpretation

This mirrors the scale-dependence already observed in H1b (Section 4): the TTN’s advantage is present at both scales but becomes stronger and more uniformly significant as network size grows. The result confirms that the one-step predictive advantage is not an artifact specific to single-step prediction — it transfers to autoregressive use without any task-specific retraining, and does so robustly across topologies. This does not yet demonstrate anything about real biological dynamical phenomena (attractors, bifurcations) — it establishes the narrower, necessary precondition that the underlying one-step model is usable in a multi-step setting at all.

9. Real-data attempt: hierarchy source matters more than noise or sample size

9.1 Motivation

All prior results use DREAM4, a simulated benchmark generated with strong modular structure by construction. We attempt to extend the predictive advantage to real biological measurement: yeast cell-cycle expression (Spellman et al., 1998, alpha-factor synchronization, 18 timepoints) aligned to a real protein-protein interaction network (STRING v12.0, *S. cerevisiae*), restricted to the intersection of genes with complete expression data and network connectivity (451 genes, later 223 for the KEGG condition below).

9.2 Attempt 1: statistical hierarchy (Louvain)

Parameter matching required extending the GNN hidden-dimension search space substantially beyond what sufficed for DREAM4 (up to 4096, vs. 256), since the TTN’s parameter count scales with network depth in a way the GNN’s shared-weight architecture does not. Once matched (6.1% difference), the experiment itself was inconclusive in an earlier attempt due to this scaling gap, and once resolved, the Louvain-based hierarchy produced 81 communities with a mean size of 5.5 genes in a 445-gene network — far more fragmented than any DREAM4 configuration.

9.3 Attempt 2: biological hierarchy (KEGG pathways)

We tested whether the problem was the *choice* of hierarchy-defining method, not real data itself, by replacing Louvain with KEGG pathway membership — a curated, biologically motivated structure rather than a statistical one. After restricting to the 223 genes (of 451) with at least one annotated KEGG pathway (pre-registered exclusion, decided after observing this fraction but before running any model), we ran the full H1/H1b protocol.

Result: H5 is falsified, in the direction opposite to the working hypothesis. The GNN outperforms the TTN with strong significance on both metrics (MSE-per-parameter: $p=3.8 \times 10^{-6}$; long-range correlation: $p=5.5 \times 10^{-5}$). Biological hierarchy did not rescue the advantage; if anything, the TTN performed worse than with the statistical hierarchy.

9.4 Isolating the cause: a factorial synthetic experiment (H6)

The two leading hypotheses for why both real-data attempts failed were measurement noise (real microarray data is noisier than GeneNetWeaver’s simulated data) and small training sample size (10 training pairs from the single Spellman time series, vs. hundreds of pairs across multiple trajectories in DREAM4). We test both, isolated, using synthetic data with a coupling structure that is *ground truth by construction*: a block-diagonal linear-dynamical system with saturating nonlinearity ($x[t+1] = x[t] + dt \cdot (-\text{decay} \cdot x[t] + \tanh(A \cdot x[t])) + \text{noise}$), where A has strong within-community coupling and weak between-community coupling, and the community partition used to construct A is identical to the TTN’s hierarchy — eliminating any uncertainty about whether the “correct” structure was supplied.

A 2×2 factorial design varies noise level (low: $\sigma=0.01$; high: $\sigma=0.3$) and training sample size (small: 10 pairs; large: 200 pairs), using the same H1/H1b protocol and parameter-matching methodology throughout.

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Condition	Noise σ	Train pairs	TTN MSE/param	GNN MSE/param	p
low noise / large sample	0.01	200	7.68e-5	5.44e-4	1.9×10^{-6}
low noise / small sample	0.01	10	2.89e-4	3.51e-4	3.6×10^{-2}
high noise / large sample	0.3	200	9.24e-4	1.33e-3	1.9×10^{-6}
high noise / small sample	0.3	10	8.51e-4	9.43e-4	1.5×10^{-2}

The TTN wins in all 4 conditions, including the one designed to replicate the adverse regime of the real-data attempts (high noise, small sample). This rules out noise and sample size, isolated or combined, as the cause of the real-data failure.

9.5 A third attempt: dynamics-derived hierarchy (also negative)

Section 9.4's synthetic result suggested a specific next step: test a hierarchy derived from dynamic information directly, rather than static topology or annotation. We implemented this using a lag-1 cross-correlation matrix ($C[i,j] = \text{Pearson correlation between gene } i\text{'s expression at time } t \text{ and gene } j\text{'s at } t+1$), with agglomerative clustering (Ward linkage, chosen after a pre-training diagnostic showed average-linkage produced a single dominant 217-gene cluster — a preprocessing decision made before observing any H1/H1b-style result) to define the partition.

This also fails, in the same direction and with similar magnitude to the Louvain and KEGG attempts: the GNN outperforms the TTN with strong significance (MSE-per-parameter: $p=9.5 \times 10^{-6}$; long-range correlation: $p=6.9 \times 10^{-4}$).

9.6 Honest reassessment

Three independent, genuinely different hierarchy sources — statistical topology (Louvain), curated biological function (KEGG), and direct dynamic coupling (lag correlation) — all fail on the same real dataset, while the synthetic factorial experiment (Section 9.4) shows the TTN winning robustly even under matched adversarial noise and sample-size conditions. This weakens our earlier interpretation that the gap was specifically about static vs. dynamic hierarchy sources. We identified three remaining candidate explanations and tested the most directly isolable one next.

9.7 A fourth attempt: trajectory structure (also ruled out)

The real series is a single trajectory split into contiguous pseudo-folds (train/val/test as temporal segments), structurally unlike the multiple independent trajectories used in both DREAM4 and the synthetic experiment of Section 9.4. We test whether this structural difference, by itself, explains the gap: using the same synthetic generator (ground-truth hierarchy, low noise — the most favorable case), we compare a multiple-independent-trajectory condition (10 trajectories $\times$ 20 steps, matching Section 9.4) against a single-long-trajectory condition (one 334-step trajectory split 60/20/20 into contiguous train/val/test segments, calibrated to yield the same 200 training pairs as the multi-trajectory condition).

The TTN wins in both conditions, at the same maximal significance ($p=1.9 \times 10^{-6}$ in each). Trajectory structure, isolated from every other factor, does not explain the gap.

9.8 Where this leaves us (after H7 and H8)

Of the three candidate explanations identified in Section 9.6, two have now been systematically ruled out by controlled experiment: (1) noise and sample size (Section 9.4) and (2) trajectory structure (Section 9.7). The remaining candidate — that real yeast cell-cycle dynamics may not have a cleanly recoverable, quasi-linear, modular coupling structure of the kind built into our synthetic generative model by construction — is the most biologically substantive possibility. We attempt to test a version of this hypothesis in a different biological system where the modular structure is better established.

9.9 A fifth attempt: *E. coli* SOS system with experimentally established module structure

The yeast cell-cycle dataset lacks an independent ground truth for whether the genes have a genuinely recoverable modular coupling structure. The *E. coli* SOS DNA damage response is different: after 30+ years of molecular biology,

two functional modules are well established experimentally — a DNA repair module (recA, lexA, recF, dinI, umuDC) and a stress sigma factor module (rpoD, rpoH, rpoS), with ssb as a cross-module connector. We use the SOS2 dataset (Kotiang & Eslami, 2020; 466 expression profiles $\times$ 9 genes, from the wpLogicNet repository) with this hierarchy fixed before any model was run.

Methodological adaptation: SOS2 contains profiles from diverse experimental conditions, not a temporal trajectory — there is no genuine $t \rightarrow t+1$ pairing. We adapt by pairing shuffled profiles cross-condition (shuffled with a fixed seed, documented as a limitation) and evaluating MSE-per-parameter and long-range correlation as before. Parameter matching yields near-perfect parity (TTN: 71 parameters, GNN: 73, 2.7% difference).

Result: mixed, with a previously unseen pattern. H1 (MSE-per-parameter) fails, with the GNN $6.7\times$ better (TTN: 0.677, GNN: 0.102, $p=1.9\times 10^{-6}$). H1b (long-range correlation) passes at extraordinary significance — but the direction reveals something specific: TTN long-range correlation is +0.224, while GNN long-range correlation is **-0.447** ($p=6.8\times 10^{-140}$). The GNN’s predictions for gene pairs from different modules are systematically anti-correlated with their true differences. The TTN captures the correct sign of cross-module relationships, while the GNN inverts them.

Interpretation: this pattern is qualitatively different from every prior real-data attempt, where the GNN dominated on both metrics. Here, the GNN learns to reconstruct individual gene profiles accurately (low MSE) at the cost of systematically inverting cross-module relationships — a failure mode that is not visible in single-gene MSE but is exposed by the long-range correlation metric. The TTN, constrained by the SOS module structure, avoids this inversion at the cost of higher individual-gene MSE. This is consistent with the TTN “spending” its representational capacity on cross-module structure rather than within-module accuracy — a tradeoff that H1/H1b in Section 4 was designed to expose and that the cross-condition pairing may exacerbate relative to a genuine temporal signal.

Whether this constitutes evidence for or against hypothesis 3 (that real biological dynamics lack recoverable modular structure) is ambiguous: the SOS result suggests that the modular structure is present and being captured by the TTN’s long-range metric, but not strongly enough to overcome the MSE disadvantage. The remaining open question is whether a temporal SOS dataset (rather than cross-condition profiling) would produce the same H1/H1b split, or whether the cross-condition pairing is suppressing the TTN’s predictive advantage in the same way we showed noise and trajectory structure do not.

10. Limitations

- DREAM4 is a simulated benchmark (GeneNetWeaver), not direct experimental measurement. All results are benchmark performance, not claims about real biological systems.
- Only two network sizes tested (10, 100 genes); scaling beyond 100 genes is constrained by the parameter-matching methodology.
- A recurrent baseline (LSTM/GRU) has not been tested.
- Bond entropy canonicalization is limited to $\text{bond_dim} \leq 4$ in the current implementation, and canonicalizes only the community subtree, not the full network.
- The DREAM4 topology inference task (Challenge 2) is distinct from the dynamics prediction task tested here.
- The synthetic ground-truth experiments (Sections 9.4, 9.7) use a simple linear-dynamical generative model (block-diagonal coupling with saturating nonlinearity); whether the conclusion (the remaining gap is likely about the structure of real dynamics, not protocol) holds for more complex, nonlinear, or stochastic generative processes is untested.
- We have not tested whether a true full Granger-causality hierarchy (multi-lag regression with F-tests, rather than the simpler lag-1 correlation clustering used in Section 9.5) would perform differently; the lag-correlation approach was a deliberate compromise given the small sample size, and rigorous Granger causality was judged infeasible at this scale, not tested and found wanting.
- The leading remaining hypothesis (Section 9.8) — that real yeast cell-cycle dynamics may lack a cleanly recoverable modular coupling structure — has not been tested directly, since doing so would require independent ground-truth knowledge of the real system’s dynamic structure, which is exactly what is unknown.

11. Conclusion

When a gene regulatory network’s community structure is known and encoded into a TTN’s hierarchy, the TTN consistently outperforms a parameter-matched GNN at predicting expression dynamics — in every one of 10 tested

configurations, against two different baselines, at maximum statistical significance for the sample size used. The advantage is not marginal: the TTN’s long-range correlation is stable across network sizes while the GNN’s degrades, pointing to a structural property of message passing (flattening of hierarchical correlations with depth) that hierarchical contraction does not suffer from.

Beyond the predictive comparison, bond entropy — a quantity with no analog in GNN architectures — predicts perturbation sensitivity without being trained for it. The rigorous version (post-canonicalization) produces a stronger signal than the proxy, suggesting the relationship is genuine rather than an artifact of gauge ambiguity. The negative result (H3) sharpens the interpretation: the benefit is specifically about having the right structure, not about having any tree structure at all. And the multi-step rollout result (H4b) confirms that the one-step advantage is not a narrow artifact of single-step prediction. We further attempted to extend the predictive advantage to real biological data across multiple pre-registered attempts, systematically ruling out noise, sample size, and trajectory structure as causes of the gap, and finding a qualitatively new pattern in the *E. coli* SOS system: the TTN captures the correct sign of cross-module relationships where the GNN inverts them, even when failing on point-wise MSE. The picture that emerges is specific: the architecture’s advantage is well-established for temporal prediction tasks where the community structure is genuinely correct, and this advantage extends to long-range structure even in non-temporal data where it fails on reconstruction MSE — pointing toward cross-condition profiling format, rather than real dynamics being unimodular, as a possible remaining explanation.

We report all of this, including the failures and the partial results, with the same rigor applied to the positive findings. The implication for the tested regime is direct: when the dynamically relevant community structure is genuinely known, encoding it explicitly outperforms learning it implicitly, across every task tested here. The open problem is narrower than when we started; the remaining work is to find datasets where both temporal structure and experimentally validated community structure coexist.

Reproducibility

All code, the full pre-registration document (including all bugs found and corrected during development, with decisions documented before outcomes were known), and experiment configuration are available at <https://github.com/XenomorphSk/CYTOS> under the Apache 2.0 license.

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References

Marbach, D., Schaffter, T., Floreano, D., Prill, R. J., & Stolovitzky, G. (2009). The DREAM4 In-Silico Network Challenge.

Schaffter, T., Marbach, D., & Floreano, D. (2011). GeneNetWeaver: in silico benchmark generation and performance profiling of network inference methods. *Bioinformatics*, 27(16), 2263–2270.

Stolovitzky, G., Monroe, D., & Califano, A. (2007). Dialogue on reverse-engineering assessment and methods: The DREAM of high-throughput pathway inference. *Annals of the New York Academy of Sciences*, 1115, 1–22.

Vidal-Saez, R. & Marbach, D. et al. (2012). Wisdom of crowds for robust gene network inference. *Nature Methods*, 9, 796–804.